EE582 Homework 01 Report

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24 September 2025

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1 Question 1

A shunt harmonic filter (series RLC) is shown in Fig. 1(a). It is designed for a three-phase 13.8 kV, 60 Hz system to shunt the 7th harmonic current. The circuit parameters are:

- $r_L = 1\ \Omega$
- $L = 19$ mH
- $C = 8.2\ \mu$F

Step 1: Write the KVL equation:

$$V_{\text{dc}} - r_L i_L - L\frac{di_L}{dt} - v_C = 0$$

Or equivalently:

$$L\frac{di_L}{dt} + r_L i_L + v_C = V_{\text{dc}}$$

With $i_L = C\frac{dv_C}{dt}$ and $v_C = \frac{1}{C}\int i_L dt$.
Step 2: Standard second-order ODE:

$$\frac{d^2 i_L}{dt^2} + \frac{r_L}{L}\frac{di_L}{dt} + \frac{1}{LC}i_L = 0$$

Step 3: Characteristic equation:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

Step 4: Calculate parameters:

$$\omega_n = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{19\times 10^{-3}\times 8.2\times 10^{-6}}} = 2533.5 \text{ rad/s}$$
$$\zeta = \frac{r_L}{2\sqrt{\frac{L}{C}}} = \frac{1}{2\sqrt{\frac{19\times10^{-3}}{8.2\times10^{-6}}}} = 0.0104$$
$$\omega_d = \omega_n\sqrt{1-\zeta^2} = 2533.3 \text{ rad/s}$$

Step 5: Quality factor:

$$Q_{\text{factor}} = \frac{\omega_n L}{r_L} = \frac{2533.5\times 0.019}{1} = 48.136$$

Step 6: Eigenvalues:

$$\lambda = -\zeta\omega_n \pm j\omega_d = -26.316 \pm 2533.3j$$

Step 7: Reactive power:

$$Q_{\text{kVAr}} = -602.04 \text{ kVAr}$$

All calculations confirm the system is very underdamped ($\zeta = 0.0104$).

Summary of Key Results:

- Natural frequency: $\boxed{\omega_n = 2533.5 \text{ rad/s}}$
- Damping ratio: $\boxed{\zeta = 0.0104}$
- Damped frequency: $\boxed{\omega_d = 2533.3 \text{ rad/s}}$
- Quality factor: $\boxed{Q = 48.136}$
- Eigenvalues: $\boxed{-26.316 \pm 2533.3j}$
- Reactive power: $\boxed{-602.04 \text{ kVAr}}$

2 Question 2

Consider the series RLC circuit with parameters r_L, L, and C as in Q1. The equations are:

KVL:

$$V_{\mathrm{dc}} - r_L i_L - L\frac{di_L}{dt} - v_C = 0 \tag{1}$$

where i_L is the inductor current and v_C is the capacitor voltage.

Capacitor current:

$$i_L = C\frac{dv_C}{dt} \tag{2}$$

Substitute (2) into (1):

$$V_{\mathrm{dc}} - r_L C\frac{dv_C}{dt} - LC\frac{d^2v_C}{dt^2} - v_C = 0$$

Rearrange to standard form:

$$LC\frac{d^2v_C}{dt^2} + r_L C\frac{dv_C}{dt} + v_C = V_{\mathrm{dc}}$$

Divide both sides by LC:

$$\frac{d^2v_C}{dt^2} + \frac{r_L}{L}\frac{dv_C}{dt} + \frac{1}{LC}v_C = \frac{V_{\mathrm{dc}}}{LC}$$

This is a standard second-order nonhomogeneous ODE. The homogeneous solution is:

$$\frac{d^2v_C}{dt^2} + \frac{r_L}{L}\frac{dv_C}{dt} + \frac{1}{LC}v_C = 0$$

Eigenvalue computation: The characteristic equation is:

$$s^2 + \frac{r_L}{L}s + \frac{1}{LC} = 0$$

The eigenvalues are:

$$s = \frac{-\frac{r_L}{L} \pm \sqrt{\left(\frac{r_L}{L}\right)^2 - 4\cdot\frac{1}{LC}}}{2}$$

For the given values ($r_L = 1\ \Omega$, $L = 19$ mH, $C = 8.2\ \mu$F):

$$\frac{r_L}{L} = \frac{1}{0.019} = 52.63\ \mathrm{s}^{-1}$$
$$\frac{1}{LC} = \frac{1}{0.019\times 8.2\times 10^{-6}} = 6,430,041\ \mathrm{s}^{-2}$$

Plug into the quadratic formula:

$$s = \frac{-52.63 \pm \sqrt{(52.63)^2 - 4\times 6,430,041}}{2}$$

The discriminant is negative, so the roots are complex conjugates:

$$s = -26.316 \pm j\,2533.3$$

Interpretation: The eigenvalues are complex conjugates with negative real parts, confirming the circuit is underdamped. Based on the very low value of the damping ratio $\zeta = 0.0104$ calculated in Q1, the transient response will exhibit oscillatory behavior with a slow decay.

General solution:

$$v_C(t) = v_{C,\infty} + [v_{C,0} - v_{C,\infty}] \left[1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t + \phi)\right]$$

where $v_{C,\infty}$ is the steady-state value, $v_{C,0}$ is the initial value, and $\omega_d = \omega_n\sqrt{1-\zeta^2}$.

For a step input, $v_{C,\infty} = x_{SS}$, and the solution simplifies to:

$$\boxed{x(t) = x_{SS} \left[1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t + \phi)\right]}$$

For the given values ($r_L = 1\ \Omega$, $L = 19$ mH, $C = 8.2\ \mu$F), the numeric solution is:

$$\boxed{x(t)[kV] = 11.3 * \left[1 - 1.00005\, e^{-26.34\, t}\ \sin(2533.3\, t + 1.5604)\right]}$$

3 Question 3

Develop a Simulink model of the circuit shown in Fig. 1(a) using conventional Simulink blocks (i.e., summers, gains, integrals, etc.). Compute the transient response (solutions) on time interval $[0, 0.1]$ s using the ode3 solver with 0.1 ms time-step size.

Block Diagram

The following block diagram represents the Simulink implementation of the RLC circuit for part (a):

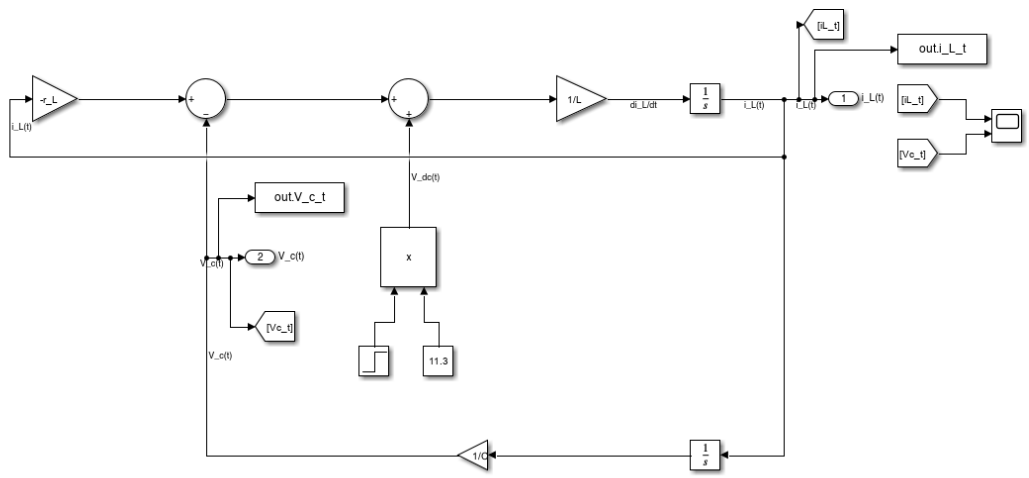

Figure 1: Simulink block diagram for the series RLC circuit (part a).

Simulation Results

The transient response of the circuit was simulated in Simulink using the `ode3` solver with a fixed time step of 0.1 ms over the interval $[0, 0.1]$ s. The following plot shows the inductor current $i_L(t)$ and capacitor voltage $v_C(t)$:

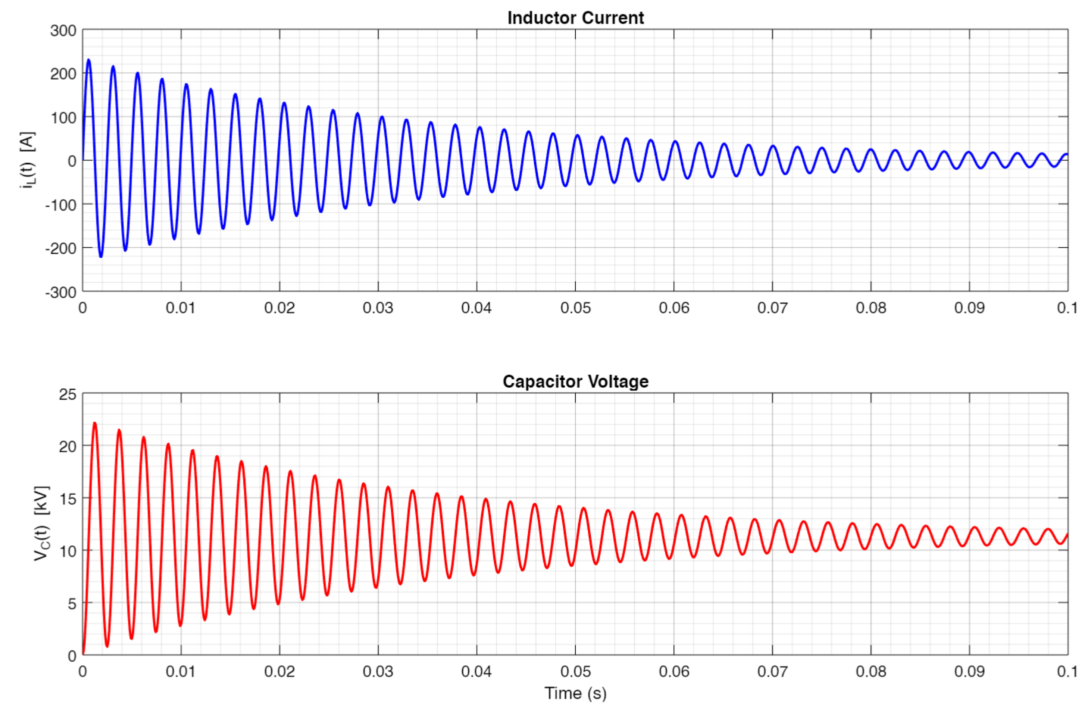

Figure 2: Simulink simulation results: $i_L(t)$ and $v_C(t)$ for the RLC circuit.

Discussion

The simulation results show the expected underdamped oscillatory response, consistent with the analytical solution derived in Q1 and Q2. The current and voltage waveforms exhibit oscillations with a slow decay, as predicted by the low damping ratio ($\zeta = 0.0104$). The Simulink model accurately captures the circuit dynamics using only basic blocks and the specified solver settings.

4 Question 4

Question:
Find the system eigenvalues using the MATLAB functions `linmod` and `eig` on the command line. Compare the results with part 2.

MATLAB Output

The following figure shows the eigenvalues computed using MATLAB's `linmod` and `eig` functions:

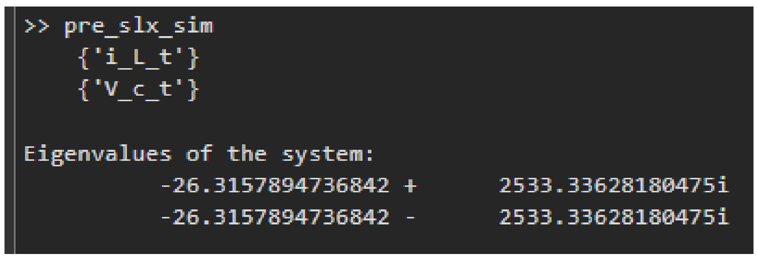

Figure 3: Eigenvalues of the system computed using `linmod` and `eig` in MATLAB.

Comparison with Analytical Results

The eigenvalues obtained from MATLAB are:

$$-26.3158 \pm 2533.34i$$

These match very closely with the hand-computed values in Q2:

$$-26.316 \pm 2533.3i$$

The small numerical difference is due to rounding and floating-point precision in MATLAB. Both methods confirm the system is underdamped with complex conjugate eigenvalues, as expected from the analytical derivation.

5 Question 5

6 Question 6

7 Question 7

8 Question 8

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12 Question 12